



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 3A: Latches Revisited

This Module, You Should...



- Differentiate between **combinational** and **sequential** designs/circuits
- Construct a synthesizable VHDL **process** to describe **sequential** circuits
- Construct sequential counters using synthesizable VHDL
- Design sequential circuits to meet lab 3 requirements

Latch Avoidance

- Avoid latching when you don't want it (in this course, we never want a latch!)

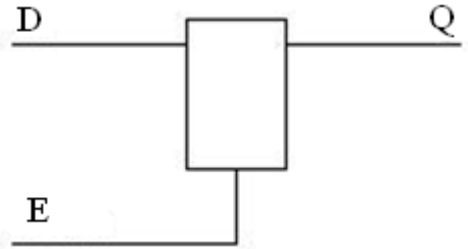
```
show_noLatch : process(pushButtons)
begin
    if(pushButtons = "01") then
        an_i <= "0010";
    elsif(pushButtons = "10") then
        an_i <= "0011";
    elsif(pushButtons = "11") then
        an_i <= "1111";
    else
        an_i <= "0001";
    end if;
end process;
```

Latch (Usually Unintentional)

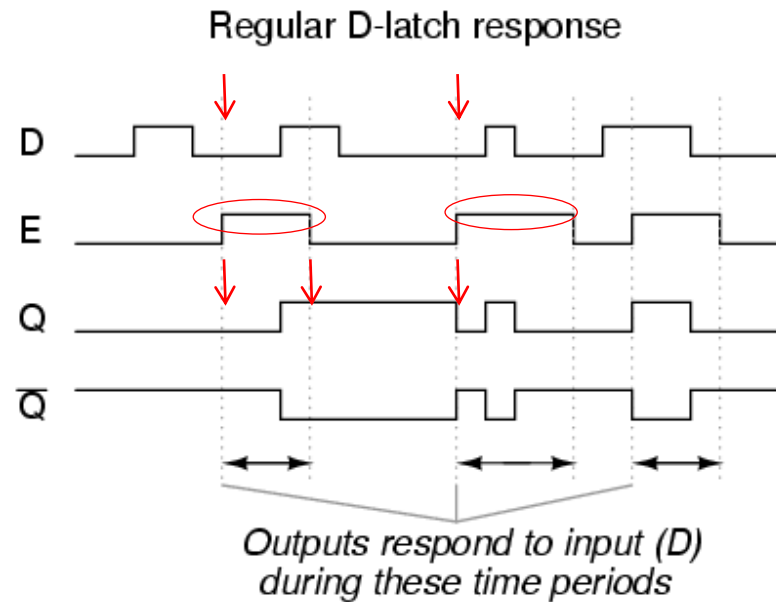
- Latching in the if/then syntax occurs when the conditional does not end with an else

```
show_latch : process(pushButtons)
begin
    if(pushButtons = "01") then
        an_i <= "0010";
    elsif(pushButtons = "10") then
        an_i <= "0011";
    elsif(pushButtons = "11") then
        an_i <= "1111";
    else
        an_i <= "0001";
    end if;
end process;
```
- When this is synthesized:
 - WARNING: [Synth 8-327] inferring latch for variable 'an_i_reg'
[D:/Projects/FPGA/Module2F_Latches.srscs/lecture2e_top.vhd:46]

Latch Inference (D-Latch)

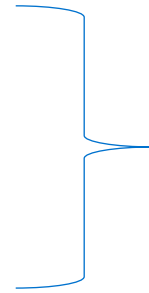


```
process (D, E)
begin
    if (E = '1') then
        Q <= D;
    end if;
end process;
```



Illustrating a Latch Inference

```
process(s, a, b)
begin
    if(s = "00") then
        c <= a;
    elsif(s = "01") then
        c <= b;
    end if;
end process;
```



Does not handle all conditions, should have unconditional else!

Synthesis Results for Latch

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Latchvsnolatch is
    Port ( a,b: in STD_LOGIC;
          s  : in STD_LOGIC_VECTOR(1 downto 0);
          c  : out STD_LOGIC);
end Latchvsnolatch;

architecture Behavioral of Latchvsnolatch is
begin
    process(s, a, b)
    begin
        if(s = "00") then
            c <= a;
        elsif(s = "01") then
            c <= b;
        end if;
    end process;
end Behavioral;
```

Utilization – Synth Report

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	1	0	0	63400	<0.01
LUT as Logic	1	0	0	63400	<0.01
LUT as Memory	0	0	0	19000	0.00
Slice Registers	1	0	0	126800	<0.01
Register as Flip Flop	0	0	0	126800	0.00
Register as Latch	1	0	0	126800	<0.01
F7 Muxes	0	0	0	31700	0.00
F8 Muxes	0	0	0	15850	0.00